Midterm A4 sheet · Data-Driven Decision Science

Side 1: how to answer each part · Side 2: calculation recipes and traps. Formulas are printed on the exam; this sheet holds what is NOT on it.

Every long answer = 4 things

Decision · number · course concept · limit on the claim. "I claim X (number). I do not claim Y."

4 rules: prediction is evidence, not action · accuracy is not utility · actions change the DGP · the decision maker owns the loss function.

Part A · concept checks (12 p, ~10 min)

- The right option usually: checks **timing** (leakage), uses **validation** then test once, compares with the **baseline**, asks for **costs**, tests on **recent unseen data incl. missing groups**, prefers the **simpler** model at equal performance.
- Score jumps 88% → 99% after adding a field → leakage. "Missing outcome" = unobserved, never "no event".
- T/F with "always / never / proves" → **False** + one counter-example from the scenario.

Part B · regression (34 p, ~35 min)

Probability from a logit

z = b0 + b1·x1 + …(dummy Yes = 1) → e^z → p = e^z/(1+e^z). Show every line; check 0 < p < 1.

Odds ratio sentence

"Each additional *[unit]* multiplies the **odds** of *[event]* by e^b ≈ __ (≈ __% higher/lower odds), **holding [other inputs] fixed**." Not probability points. Association.

"Bigger coefficient = more important"?

Not fair: different units (0/1 dummy vs per unit / per SD). Compare over a realistic change (k units × b, or per SD). Size ≠ cause.

Linear coefficient sentence

"Among [range of data], each extra [unit of X] is associated with [b] more [units of Y] on average, holding [others] fixed."

p-value vs 95% CI

CI = b ± 2·SE (exam allows ±2). CI gives **size and precision** (plausible range, convert to OR with e^); p only says "unlikely zero". Includes 0 → not significant.

Duplicated rows / huge n: SE shrinks artificially (no new info, independence broken) · huge n makes tiny effects "significant" → statistical ≠ practical.

"p < 0.001 so doing X will reduce Y" → Not supported

Association ≠ intervention; confounding (marker of sicker / more motivated people); risk ≠ persuadable.

A/B test template: Who: eligible [people], **randomly assigned** to [action] vs usual care. **Compare:** [outcome rate] (and cost) between groups. **Why random:** groups alike on everything, measured or not → difference is caused by the action. **Check:** baseline balance · outcome recorded for both groups (incl. elsewhere) · action actually delivered · enough sample · outcome fixed in advance.

Group missing from training (e.g. nursing facility, app sign-ups)

0% in training → score is extrapolation, calibration unknown. **Limit** (don't use for them, current process) + **Learn** (shadow-score, collect outcomes, validate/calibrate).

Residual table by band

- Mean residual trends (− → +) or U-shape → relationship is **curved**; line wrong form → add squared term / bands / spline.
- Spread grows → predictions **less precise** there than average RSE says.
- Low R^2 → input explains little. Few rows in a band → weak evidence.
- Predicting outside the data range (6 bedrooms when data 1–4) → **extrapolation**, don't trust.

Collinearity (corr ~0.9)

Same information → SE inflate, signs odd, neither significant alone, fit (R^2) unchanged; sum stable. VIF = 1/(1−r^2) (2 inputs). **Prediction:** fine, keep best test model. **Explanation:** don't read separate coefficients; use one / an index / joint effect / new data.

Three models: train vs test

Big train–test gap (0.95 vs 0.66) = **overfit** (too many inputs, high variance). Recommend best **test** + small gap + explainable. Before paying for extra data: available at decision time? recorded for everyone? gain lowers cost at the threshold?

Confounding / sign flip

X alone looks risky, with Z it flips: Z drives the outcome and X-people have more Z. At the same Z, X is safer. Don't act on X alone.

Part D · audit the AI (25 p, ~20 min)

Claim pattern	Verdict + reason
Training R^2 → "explains % of next year"	No: training fit, optimistic; new data may differ
p < 0.001 → "doing more X reduces Y"	No: association, confounding; needs A/B
"X% accurate, ready"	No: compare with base rate (calling nobody = 1 − rate)
"Flags a of b who had the event"	Check table: TP = a? recall a/b → Supported if it matches
Joiners / users did better → program works	No: self-selection; randomize the invitation
"AUC 0.81 → 81% of flagged will …"	No: AUC = ranking; that is precision

Expected vs actual cost by group: trust the group with the biggest actual/expected ratio least (model under-predicts). Check: calibration by score band for that group on recent outcomes + how many were in training. If actual > predicted → **lower** that group's threshold / recalibrate up / Limit.

Memo ≤ 40 words each.
JUDGMENT: "The model ranks risk well (test AUC __). Within the __-call limit, threshold __ costs __ per 1,000, versus __ for [best feasible baseline]. It predicts risk; it does not prove [action] works."
ACTION: "From Monday, [owner] [acts on] cases scored ≥ __ (about __ per 1,000); a random 10% get usual care to measure the effect. Monitor monthly: [group]'s actual versus expected [event rate]."

Side 2 · Part C recipes, numbers, traps

Part C · cost and threshold (29 p, ~30 min)

From the counts table

Write TP/FN/FP/TN first. Check TP+FN = positives, FP+TN = negatives, flagged = TP+FP.

```
Recall = TP/(TP+FN)   Precision = TP/(TP+FP)
Specificity = TN/(TN+FP)   Accuracy = (TP+TN)/N
Cost = C_FN·FN + C_FP·FP
Do nothing  = C_FN · (all positives)
Act on all  = C_FP · (all negatives)
```

Meaning sentences: recall = "catches _% of those who [event]; misses _"; precision = "_ of every 10 [actions] go to people who really [event]"; specificity = "_% of those without the event are left alone".

Which policy?

Lowest cost **among feasible** policies. Say how much it beats each baseline. Note if the 0.50 model is worse than a simple policy.

Accuracy hides…

Base rate: doing nothing scores (1 − rate) while catching 0. Accuracy weights a miss (C_FN) like a false alarm (C_FP).

Why accuracy ↓ while cost ↓ (lower threshold)

```
ΔFN × C_FN (saved)  vs  ΔFP × C_FP (added)
e.g. −75 × 9,000 = −675,000 ; +235 × 1,000 = +235,000 → net −440,000
```

More total errors, but cheap ones replace expensive ones; rare event → catching it costs false alarms.

Is the model worth running?

```
Saving/1,000 = best baseline cost − model cost
Saving/year = saving/1,000 × (volume/1,000)
Net = saving/year − running cost ; return = saving/running
Stop when running cost > saving/year
```

Caveat: costs and calibration must hold; capacity for the flagged volume.

Break-even

p* = C_FP/(C_FP+C_FN) = 1/(1+r), r = C_FN/C_FP. Act when p > p*. Bigger miss cost → lower p*. Needs **calibrated** p.

r = 4 → .20 · 5 → .167 · 8 → .111 · 9 → .10 · 16 → .059

Capacity

- Feasible = flagged ≤ capacity. Pick the cheapest feasible.
- Value of more capacity = cost(best feasible) − cost(better infeasible) per 1,000; ÷ extra actions = max worth paying per extra action; × volume/1,000 per year.
- Other levers: add staff, triage, rank and act on the top N.

Uncertain cost range (CFO)

- If one threshold is cheapest at both ends → it **dominates**: choose it.
- Else: regret = cost − best cost in that column; pick the smallest worst-case regret.
- Underestimating the miss cost is usually the expensive mistake (curve steep where you act too little). Ask Finance to narrow the range.

ROC

- x = FPR = FP/(FP+TN), y = recall. Lower threshold → point **up and right**.
- Diagonal = coin flip (AUC .5). AUC = P(random positive scored above random negative).
- ROC can't pick the point: no costs, no base rate, no capacity, no calibration. Choose the point nearest p* that is feasible.

Calibration

"Among people scored ~30%, ~30 in 100 have the event." AUC = order, calibration = size; break-even needs size. Averages hide subgroup failure (33.6% vs 33.1% overall, first-time 2% → 28%). Resampled 50/50 → probabilities inflated: recalibrate or threshold by rank. Fix: recalibrate · Learn: pilot · Limit: only where evidence.

Data audit · 5 questions

Question	Ask	Response
Representativeness	Who is missing?	Limit + Learn
Selection	Did past decisions hide outcomes?	Learn (capped pilot)
Leakage	Known at prediction time?	Fix (drop, re-run)
Label quality	Label = outcome we care about?	Fix (redefine / validate)
Deployment fit	Use looks like test?	Limit (staged, stop rule)

Audit statement: **Observation → Mechanism → Implication → Next evidence.** More rows fix sampling error, not selection bias. Train/val/test inherit the same selection.

Bias–variance

Test error = variance + bias² + irreducible. Flexible (K=1, many inputs): low bias, high variance, train ↓ test ↑ (U-curve). Validation chooses, test once. Training R^2 never falls with inputs; use test/adjusted R^2. Data snooping: P(≥1 false hit) = 1 − 0.95^P (P=20 → 64%, 30 → 79%, 40 → 87%).

Traps checklist (scan before handing in)

- Did I show working for every number?
- Said "associated with", not "causes"?
- Compared with the baseline / base rate?
- Respected capacity?
- Named an owner, a metric and a trigger (not "monitor carefully")?
- Gave the size (CI / OR), not just "significant"?
- Checked the group the model never saw?